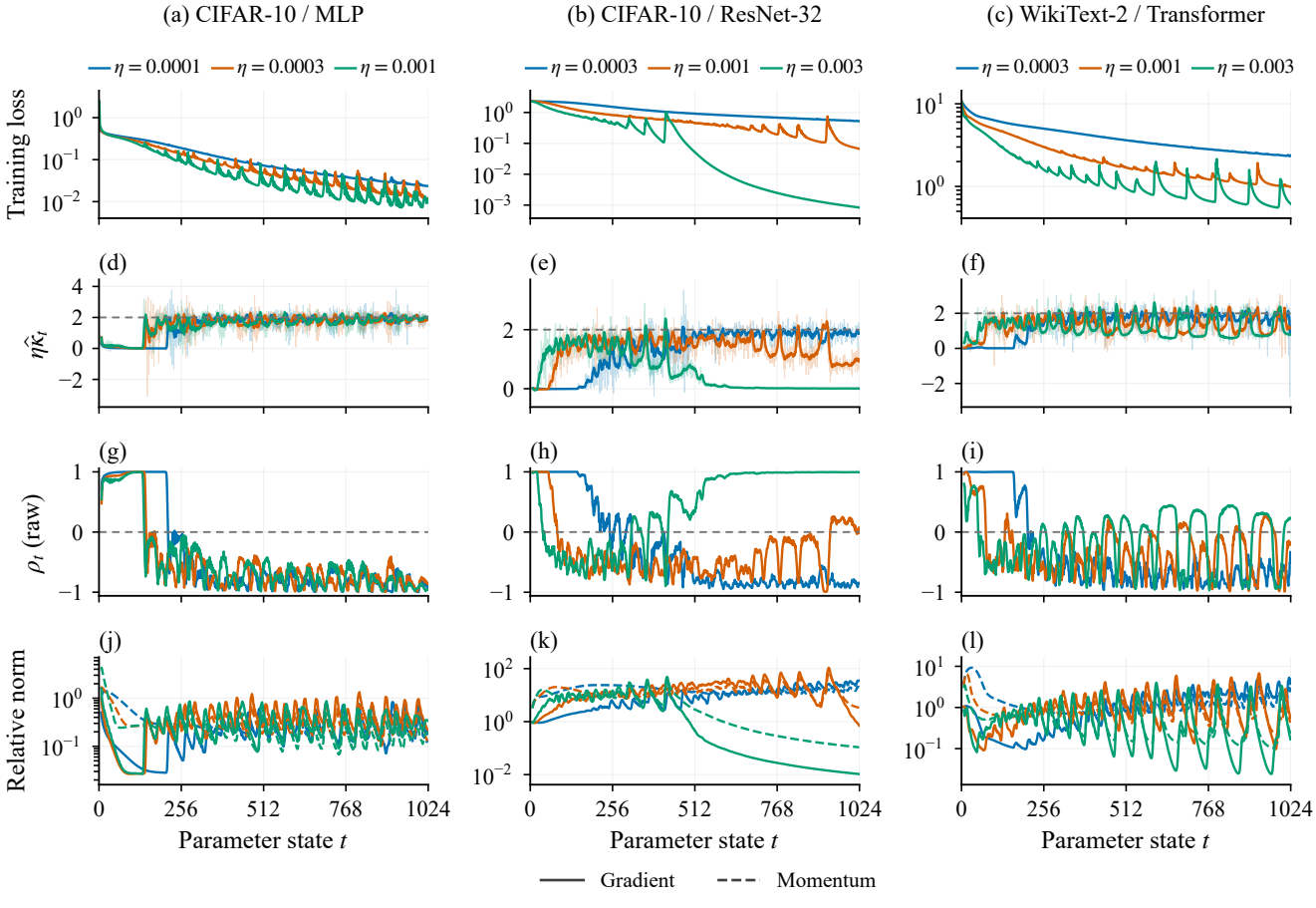

(a) CIFAR-10 / MLP
(b) CIFAR-10 / ResNet-32
(c) WikiText-2 / Transformer
η = 0.0001
η = 0.0003
η = 0.001
η = 0.003
Training loss
(d)
(e)
(f)
$\hat{\eta}\kappa_t$
(g)
(h)
(i)
ρ_t (raw)
(j)
(k)
(l)
Relative norm
0
256
512
768
1024
Parameter state t
Gradient
Momentum